

InternshipAWS

AWS Cloud Infrastructure Project

Project Report

A hands-on implementation demonstrating cloud networking, compute, storage, identity and access management, monitoring, notifications, backup, log analysis, and cost management using AWS.

Project Summary

InternshipAWS is a practical AWS cloud project in which a web application was deployed on an Ubuntu EC2 instance using Apache. The environment was supported by a custom VPC, IAM controls, S3 storage, CloudWatch monitoring, SNS alerts, EBS snapshot backup, and an AWS Budget. Web-access logs were also analyzed to derive traffic and error insights.

1. Project Objectives

- Build and deploy a working web application on AWS.
- Configure a basic public cloud network using VPC components.
- Apply IAM-based access control and least-privilege principles.
- Use S3 for project object storage.
- Monitor the environment with CloudWatch and configure SNS notifications.
- Create an EBS snapshot for backup and document restore considerations.
- Analyze web-access logs and extract useful operational metrics.
- Monitor cloud spending using an AWS Budget and follow Free Tier safety practices.

2. AWS Services Used

Service	Purpose
Amazon VPC	Network isolation, subnet and routing configuration
Amazon EC2	Ubuntu web server hosting the Apache application
Amazon S3	Project-related object storage
AWS IAM	Users, groups, roles and permission management
Amazon CloudWatch	Monitoring and EC2 alarm configuration
Amazon SNS	Email notifications for alerts
Amazon EBS	Persistent EC2 storage and snapshot backup
AWS Budgets	Cost monitoring and threshold alerting

3. Architecture & Network Design

The project uses a custom VPC containing a public subnet. An Internet Gateway provides Internet connectivity, while a route table contains the public route. The EC2 instance is deployed in the public subnet with a public IPv4 address so the Apache web application can be accessed for demonstration and verification.

Logical flow: User → Internet → Internet Gateway → Public Subnet → EC2/Apache Web Server. Supporting services include IAM, S3, CloudWatch, SNS, EBS and AWS Budgets.

4. Compute & Web Application

The web server uses an Ubuntu 24.04 LTS EC2 instance with Apache HTTP Server. The deployed application provides an InternshipAWS demonstration page and was verified through the browser and command-line HTTP request testing.

5. IAM & Security

IAM was used to manage access to AWS resources. The project followed a least-privilege approach for required permissions and used an EC2 instance role for application-level AWS access. Sensitive credentials and private key material are intentionally excluded from this public project report and GitHub repository.

6. S3 Storage

Amazon S3 was used for project-related object storage. The bucket structure and sample objects were verified as part of the project evidence.

7. Monitoring & Notifications

Amazon CloudWatch alarms were configured for EC2 monitoring, including instance status and CPU-related conditions. Amazon SNS was used to deliver email notifications, and the email subscription was confirmed.

8. Log Dataset & Analysis

A dataset containing 300 generated web-access log records was analyzed to identify traffic patterns, popular pages, errors, and unique visitor IPs.

Metric	Result
Total log records	300
Unique visitor IPs	183
Total errors	89
HTTP 404 errors	35
HTTP 500 errors	54
Peak traffic hour	00:00–00:59
Requests during peak hour	18

Most Accessed Pages

Page	Requests
/health	57
/css/style.css	53
/login	52
/index.html	51
/about.html	44

9. Backup & Cost Management

An 8 GiB EBS snapshot was created as backup evidence. The project also used an AWS Budget named **InternshipAWS-Monthly-Budget** with a monthly threshold of \$10 and an 80% alert threshold. Free Tier safety practices included keeping only the required micro EC2 instance running, avoiding unnecessary NAT Gateway, ALB and RDS resources, limiting storage and log ingestion, and maintaining a limited snapshot count.

10. Verification & Evidence

The project was verified using browser access to the deployed web application, curl testing from the EC2 environment, AWS console screenshots, monitoring alarms, SNS email notification evidence, S3 object evidence, EBS snapshot evidence, and budget configuration evidence.

11. Key Learning Outcomes

This project provided practical experience with AWS infrastructure, Linux server administration, Apache web hosting, VPC networking, IAM, object storage, monitoring, event notification, backup concepts, log analysis, and cloud cost management.

12. Project Deliverables

- Architecture diagram
- AWS implementation screenshots
- EC2 web application proof
- Web-access log dataset and analysis results
- CloudWatch and SNS monitoring evidence
- EBS snapshot and restore notes
- AWS Budget and Free Tier safety checklist
- Project explanation video
- GitHub project repository

13. Conclusion

InternshipAWS demonstrates an end-to-end, hands-on approach to deploying and operating a small cloud workload. The project combines infrastructure, security, application hosting, monitoring, backup, data analysis, and cost awareness into a single AWS implementation.

Repository: InternshipAWS

Note: This report intentionally excludes credentials, private keys, and other secrets.